9.4 Independent and Dependent Events

Name _____ Period ____ Date _____

Algebra I 9.4 · Student sheet

I can…

- tell whether two events are independent or dependent.
- find the probability of independent events.
- find the probability of dependent events.

Vocabulary · fill in the missing word

Independent events — the _____ of one event does not affect the other

Dependent events — the result of one event changes the _____ of the other

Independent — P(A and B) = _____

Dependent — P(A and B) = _____

With replacement — the item is put back, so the events are _____

Without replacement — the item is kept, so the events are _____

Part 1 · With or without?

Your bag has 3 red and 2 blue chips. Do each round 10 times. Record R for red and B for blue.

Round	1	2	3	4	5	6	7	8	9	10	Both red
With replacement											
Without replacement											

Class totals: with replacement _____ out of _____ without replacement _____ out of _____

Which round gave “both red” more often? Why?

Part 2 · Notes and examples (fill in each note, then work the example)

Independent: one event _____ the probability of the other. P(A and B) = _____

Replacement: With replacement → _____. Without replacement → _____.

Example 1

Tell whether the events are independent or dependent.

a. You roll a die and flip a coin.

b. You draw a card, don’t put it back, and draw another card.

c. A class elects a president, then elects a vice president from the other students.

Example 2

You spin a spinner with four equal sections (red, blue, yellow, green) and roll a die. What is the probability of red and a 6?

Dependent: P(A and B) = P(A) · _____

Without replacement: Without replacement, the second fraction has one fewer in the _____ and in the _____.

Example 3

A drawer has 4 black socks and 6 white socks. You pick two socks at random. Find P(both black)

a. with replacement. b. without replacement.

Example 4

Events A and B are independent. P(B) = 0.4 and P(A and B) = 0.1. Find P(A).

Part 3 · Partner practice

Partner 1

A jar has 5 red and 3 green candies. You eat one at random, then take another. What is the probability that both are green?

Partner 2

The probability of rain on Saturday is 0.3, and on Sunday it is 0.5. Assume the days are independent. Find the probability of rain on both days and the probability of no rain either day.